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Solar and Stellar Activity Cycle Forecasting with Gaussian Process Regression

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Abstract

Exoplanetary searches by the radial velocity and transit methods are affected by noise from stellar activity, which can produce significant non-planetary signal variations, degrading exoplanet detectability. Stellar activity and thus noise magnitude oscillate cyclically, implying periods during which observations are more sensitive to exoplanets. By optimising observations around these periods, oversubscribed telescopes will be able to more efficiently discover exoplanets. This work builds a scalable pipeline for the forecasting of these cycles using Mount Wilson Observatory stellar activity S-Index data such that observations looking for exoplanets can be made at activity minima. A Gaussian process regression (GPR) with a spectrum kernel, a sum of decaying quasi-periodic kernels, is used for the predictions. Each kernel mode is physically motivated to correspond to the rotation cycle, magnetic activity cycle, and long-term trends of stellar activity. Predicted best observation times were accurate to within approximately one year for benchmark stars over forecast horizons of up to five years, compared to the multi-year timescale of the underlying activity variations. Performance was evaluated on a small set of benchmark stars; a fuller evaluation on a larger stellar dataset was limited by HPC failure. Results of a synthetic-population study on the data cadence required for good predictions are reported and recommendations for further work are made.

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I would also like to thank my parents for encouraging my academic pursuits and supporting me every step of the way. Very few Malaysians have the chance to study physics, let alone at an institution like Cambridge, and I am deeply grateful for this opportunity.

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Chapter 1

Introduction

1.1 Field Background and Key Problem

The first detected exoplanet around a main-sequence star was discovered by measuring the Doppler shift from slight changes in the host star's radial velocity due to the presence of the planet. [1] Since this first detection in 1995, by a combination of the radial velocity method and the transit method, which detects planets by the dip in stellar luminosity as exoplanets occlude their host stars [2], over 6000 exoplanets have been detected. [3]

Unfortunately, neither observable, the Doppler shift nor the variation in luminosity result uniquely from exoplanetary effects. Stellar activity can induce non-exoplanetary variations in these measurements. During times of increased activity, dark stellar spots and bright plages form on stellar surfaces; this adds noise to luminosity measurements, causing greater uncertainty in the transit method of exoplanet detection. As the stellar surface rotates, the hemisphere rotating towards the detector appears blue-shifted while the hemisphere rotating away appears red-shifted. With uniform luminosity, the overall effect cancels. However, when spots and plages form, the symmetry is broken, causing an apparent Doppler shift, which can mimic and obscure true radial velocity variations. [4]

The presence of these noise-generating phenomena is not constant; they are correlated with variations in stellar activity and have a measurable effect on the observability of exoplanets. Studies comparing solar radial velocity data, as an analogue for other stars, taken during solar maxima and minima show that the induced noise at solar minimum is ~ 0.5 m/s while at solar maximum it is an order of magnitude higher. This is significant compared with the ~ 0.1 m/s signal an Earth-like planet in the habitable zone would induce on a Sun-like star. [5] Similar analysis by Sairam & Triaud shows that at exoplanet orbital periods of greater than 100 days the detection limit for exoplanets in orbital semi-amplitude can be up to two orders of magnitude lower when observing at solar minima than maxima. [4]

Presently, target stars are observed with no knowledge of where they are in their activity cycles, so detection efficiency is not maximised. This exacerbates the limitations in exoplanet detection rates caused by the oversubscription of the high-precision instruments needed for exoplanet searches: precise instrumentation is needed to isolate the Keplerian contribution to radial velocity or to detect the minuscule fractional luminosity dips from transits. As such, the limited time available on high-precision instruments could be better used by only observing cyclical stars at their stellar minima, improving the chances of exoplanet detections. Sairam & Triaud argue for a public forecast of stellar activity to enable this optimisation and build a simple sinusoidal model as a proof-of-concept. [4] Since the goal of such a forecast is to characterise activity cycles, it does not require the same instrumental precision as the exoplanet searches themselves; forecasting can instead be performed on data from more modest instrumentation.

1.2 Current State of Stellar Activity Predictions

There is much work forecasting the solar cycle using sunspot counts, since they are the longest-running proxy for solar activity, with hundreds of years of data. Gonçalves, Echer & Frigo (2020) applied a warped Gaussian process regression to annual sunspot number data from 1700–2018 to forecast cycle amplitude and timing, where the warping is used to enforce non-negative predictions. [6] When validated on holdout data, these methods achieved an RMSE of ~ 10 within a five-year forecast horizon against typical solar maximum peaks of 100–170. Given the Sun's roughly 11-year cycle, this performance suggests useful minima and maxima forecasts are achievable from the midpoint of the cycle: ample time to schedule optimised observations.

1.3 Stellar Variability

The variability in the S-Index of stars arises from a series of physical phenomena.

Stars have magnetic activity cycles originating from the periodic reorganisation of stellar magnetic fields. The Sun experiences variations from this cycle with periods fluctuating around 11 years, manifesting as clear variations in sunspot count and space weather activity. Both the amplitude and duration of this cycle vary. [7]

There is also a shorter-term cycle corresponding to the stellar rotation rate. Groups of spots or bright plages can survive for several stellar rotation periods, during which they rotate in and out of observation. [8] The presence of these phenomena causes variations in the S-Index as they have different emission characteristics when compared to the surroundings. The rotational variation magnitude varies with the magnetic activity cycle as peak magnetic activity corresponds to more spots and plages and therefore greater resulting S-Index variation from rotation.

Historical sunspot observations have exposed a further long-term trend modulating the amplitudes of the magnetic cycle variations called the Gleissberg cycle. This oscillation in peak sunspot counts happens with an approximately 88-year period. [9] Such cycles have not yet been reliably detected in other stars due to the short timespan across which stellar observations have thus far been taken.

Further signals exist on shorter timescales, such as p-mode oscillations and stellar granulation, which vary on timescales of the order of minutes and hours respectively. [4] These will not be explicitly modelled as the sampling frequencies available do not approach the Nyquist limit for their detection.

1.4 Objectives

The goal of this work is to improve on the proof-of-concept sinusoidal model by Sairam & Triaud and to design a pipeline that can produce, on a large scale, stellar activity forecasts to aid in the optimisation of exoplanet observations.

Chapter 2

Data

A considerable challenge for this project is the poor availability of long-term, regularly sampled stellar activity data. For the Sun, activity has been observed by sunspot counts for centuries, but this is not possible for other stars.

As such, the S-Index, a spectroscopic proxy for stellar activity, is used. This is calculated using the flux scales of the Ca II H and K emission lines by

$$S = \alpha \frac{H + K}{R + V}, \tag{2.1}$$

where H and K are the fluxes integrated over narrow passbands around the Ca II H and K line centres, and R and V are the continuum bands around them. α is a calibration factor that is calculated by measurements of standard lamps and stars. [10, 11]

This index was chosen because there is a publicly available dataset of S-Index measurements from an observing campaign of the Mount Wilson Observatory (MWO) including thousands of stars. These were observed between 1966 and 1995. [12] The index is also ideal in that it can be calculated from spectroscopic data from modest instrumentation, making this work applicable beyond the regularly sampled spectra in the MWO dataset.

MWO also made sun-as-a-star observations: observing the Sun as though it were any other star and producing S-Index measurements for it. The availability of cycle predictions made using sunspot data in the literature yields a valuable baseline against which these S-Index predictions can be verified.

The sampling cadence and data timespan vary considerably. Some stars, such as the four plotted in Fig. 2.1, are clearly cyclical and their behaviour is well characterised in the literature. As such, they will form a baseline set of stars against which forecasts can be compared. The majority of the observed stars, such as the one plotted in Fig. 2.2, are much more sparsely sampled. It will be difficult to produce predictions for these stars, so the focus will be on the well-sampled stars and then on the generalisation of the modelling to more stars.

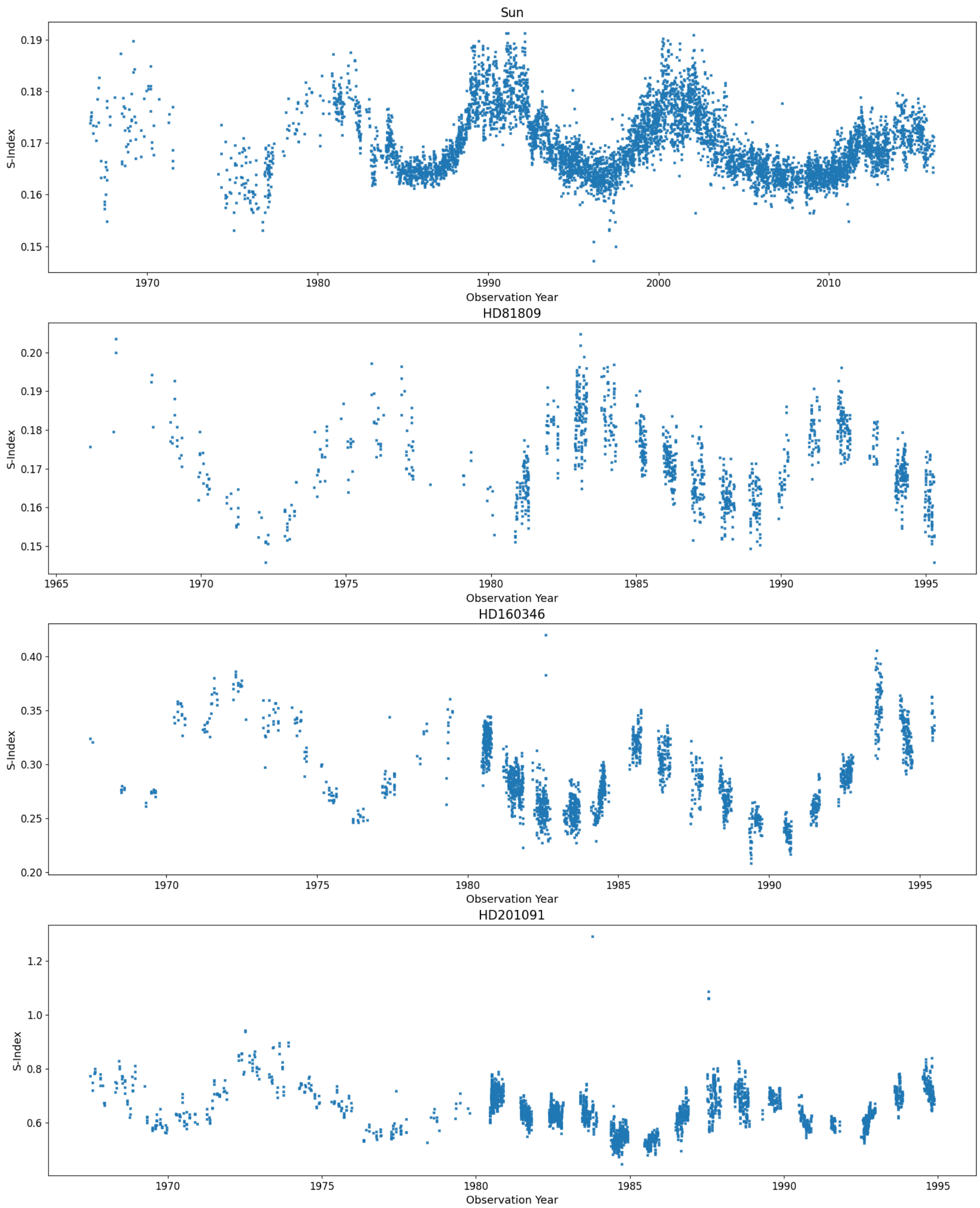

Figure 2.1: The S-Index time series for four stars: the Sun, HD81809, HD160346, and HD201091. These are examples of well-sampled MWO S-Index data. The observations show clear cyclical behaviour and are seasonally sampled. The solar data shows a clear increase in variance at activity peaks; this is physically motivated as there will be greater rotational variation since magnetic cycle peaks are associated with greater numbers of sunspots.

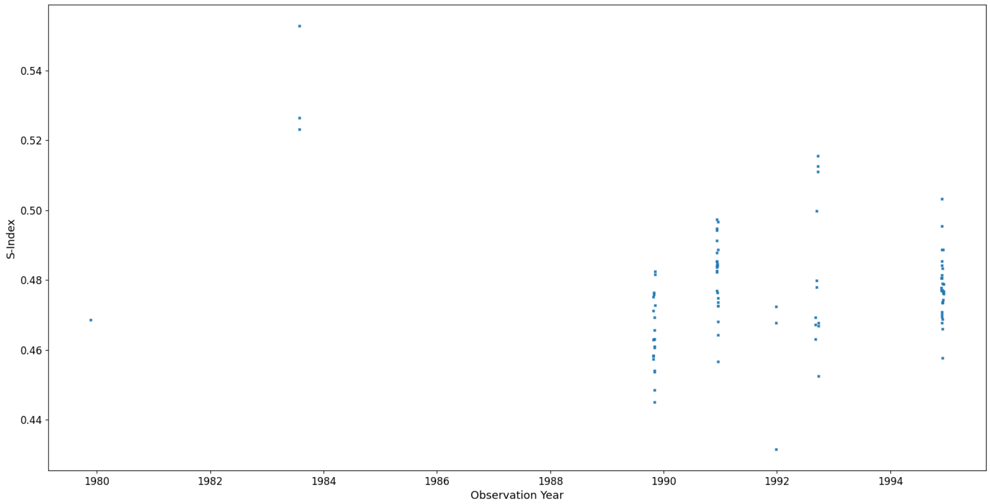

Figure 2.2: The S-Index time series for HD166, an example of a poorly sampled star from the MWO dataset. There are few sampled epochs and large gaps between observations.

The MWO dataset does not provide uncertainties for the S-Index measurements. Baliunas et al. (1995) estimated the uncertainty on the data to be between one and two percent. [7] To be conservative given the range in data quality, the uncertainty was set to five percent.

The data was prepared by dropping rows with NaN values. Furthermore, the data occasionally exhibited outliers; some can be seen in the HD201091 time series in Fig. 2.1, where three points exist far outside the previously observed ranges. It is important to remove these as they will distort the predictions, especially for sparsely-sampled datasets. To identify these outliers dynamically for every dataset, accounting for the natural differences in the variability of the data, the median absolute deviation (MAD) from the median of the S-Index was taken for each star. Data points with a deviation from the median of more than three times this MAD were removed as outliers.

Chapter 3

ARIMA

3.1 ARIMA Modelling

Before applying Gaussian process regression methods to the S-Index data, ARIMA, a well-understood time series model, was applied to establish a performance baseline.

ARIMA models have three components. Autoregression (AR) predicts on the previous p data points

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \epsilon_t. \tag{3.1}$$

The moving average (MA) term uses the previous q prediction errors

$$y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + ... + \theta_q \epsilon_{t-q}. \tag{3.2}$$

ARIMA models need stationary data: time-invariant mean and variance. This is enforced by the integration (I), which differences the series to order d. [13]

The corresponding points in previous cycles can also be considered to explicitly model seasonality:

$$y_t = \text{ARIMA} + \phi_1 y_{t-m} + \phi_2 y_{t-2m} + \cdots + \phi_p y_{t-pm}, \tag{3.3}$$

where m is the seasonal period, with an analogous term included for seasonal MA.

3.2 ARIMA Data

A difficulty in applying ARIMA is that it requires regularly sampled data, which is hard to come by in astronomy. Due to seasonal visibility, stars cannot always be observed. Furthermore, scientific priorities and funding change, so long-term observing campaigns may not be continuous.

The data was downsampled in regions of abundant data and linearly interpolated in regions without data. A limit of five interpolated data points was set to prevent over-reliance on non-physical data. If more than five interpolated points were required to fill any gap, the downsampling of that dataset to that cadence was deemed unsuitable and a coarser time resolution was used.

3.3 Solar ARIMA

In the solar time series plotted in Fig. 2.1, there is a prolonged break in observations around 1972. The smallest sampling interval that crosses that gap without excessive interpolation is semi-annual. As such, ARIMA analysis was performed on the Sun with the annual, semi-annual, and post-gap monthly samplings plotted in Fig. 3.1.

To verify data stationarity, the autocorrelation (ACF) and the partial autocorrelation (PACF) were plotted in Fig. 3.2. ACF measures the correlation between the series and its lagged values; the PACF measures the correlation at a given lag after removing the effect of intermediate lags. The shaded 95% confidence interval represents the bounds within which autocorrelations are statistically indistinguishable from zero under noise fluctuations.

The confidence interval quickly grows to eclipse the autocorrelations due to the short data span as there are only four cycles of solar data. The ACF pattern corresponds to what is expected given the 11-year solar activity cycle: points are highly correlated with the 11-year lags and anti-correlated with the 5.5-year lags. The autocorrelation magnitude decreases with increased lag; this is expected as the length of the solar magnetic activity cycle varies and is not exactly 11 years, causing longer lags to drift out of phase and decorrelate. Such an autocorrelation pattern implies data stationarity as non-stationary data exhibits a slow ACF decay.

The autocorrelation implies a weak MA term: a q-th order MA exhibits an ACF cut-off at lag q. The PACF is strong for small lags then falls within the interval quickly, which is indicative of a low-order AR term.

Because a low-order AR term does not cover the full cycle, it cannot capture the seasonality without seasonal ARIMA terms. The value of m was set to correspond to 11 years. This is problematic for finer time resolutions because calculating a seasonal ARIMA is O(m^2). With monthly data, m is 132 and this becomes intractable. As a result, further analysis was performed with just the annual and semi-annual cadences. It is also worth noting that, because the cycle period itself varies, a fixed value of m cannot fully capture the underlying dynamics.

The ARIMA was implemented using `pmdarima`. [14] Using Fig. 3.2 to inform initial p and q guesses, the Hyndman-Khandakar algorithm was used to find the optimal hyperparameters. [15] The loss function was the corrected Akaike information criterion (AICc) as it combines likelihood with an Occam penalty term and corrects for small training

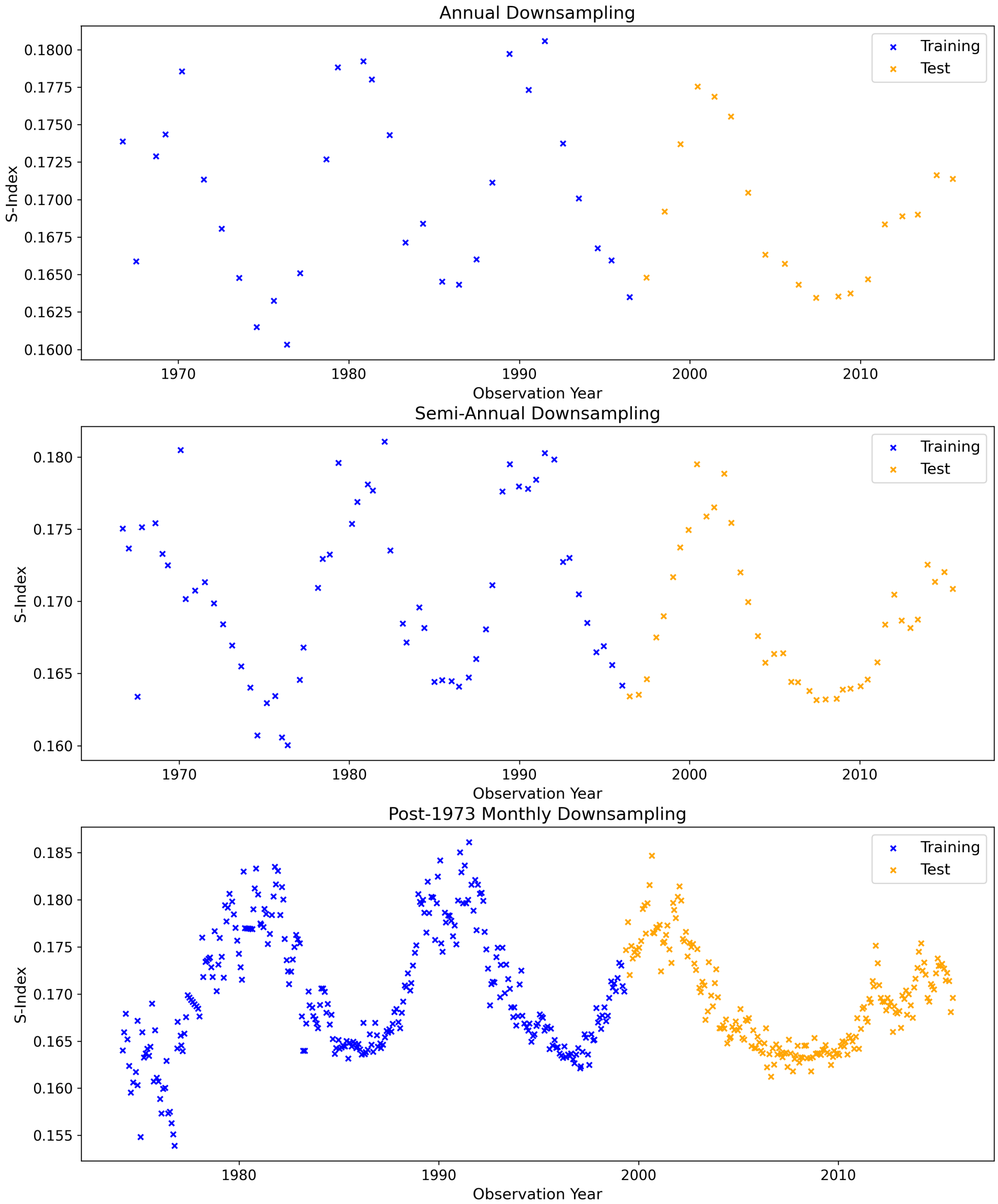

Figure 3.1: The three downsampling cadences of the solar S-Index examined. In the semi-annual and monthly cadences, there are signs of interpolation where subsequent points trend linearly.

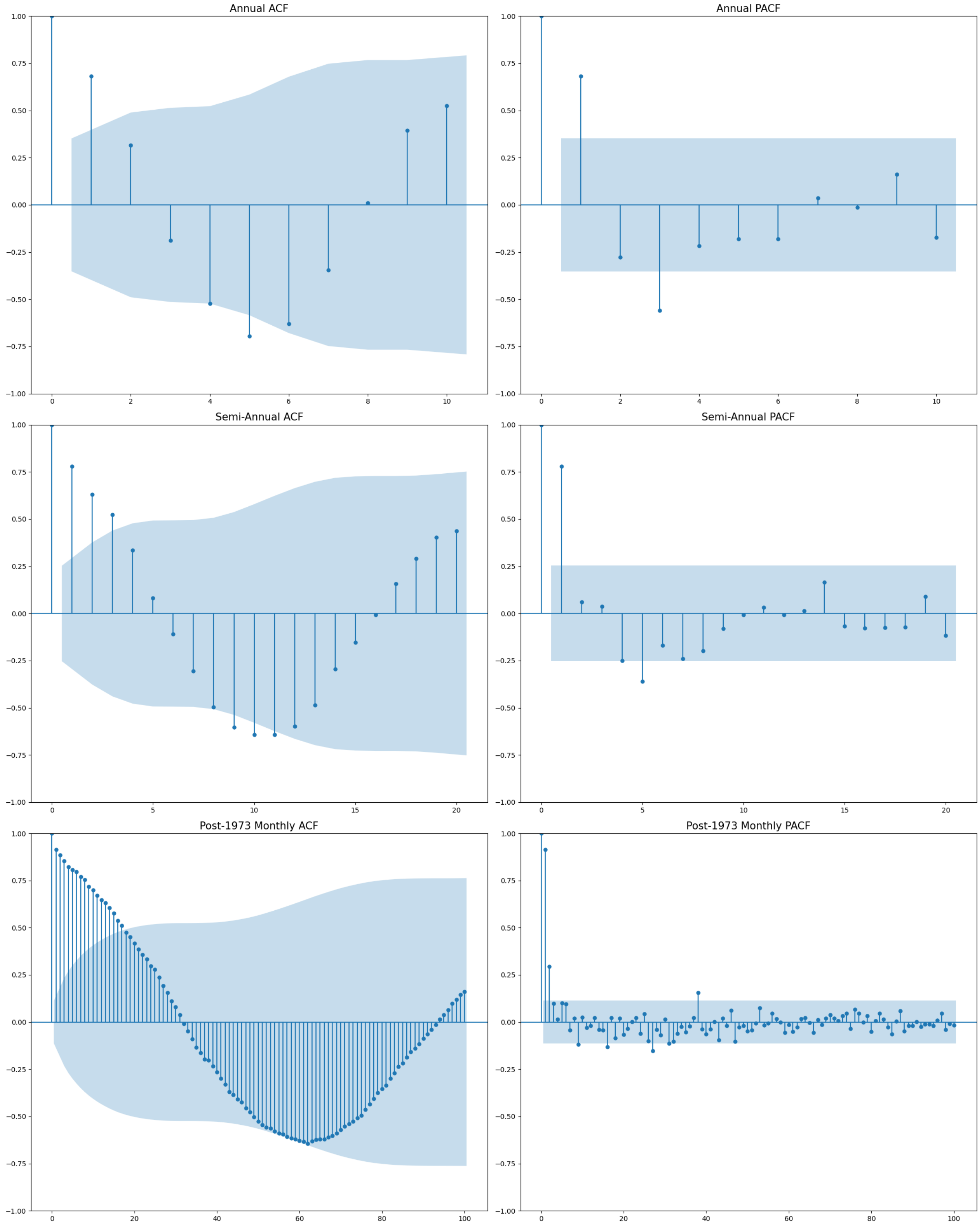

Figure 3.2: ACF and PACF for various lags in the solar data with 95% noise confidence interval (blue filled). The ACFs indicate data stationarity and exhibit oscillatory behaviour consistent with the Sun's 11-year solar cycle.

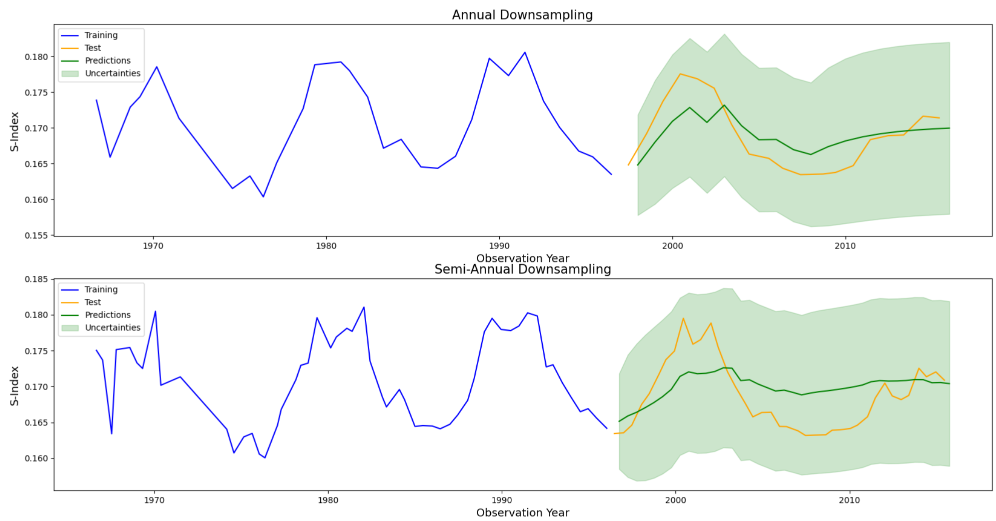

Figure 3.3: The predictions of the ARIMA models for the annual and semi-annual samplings plotted with the training and test data. Both follow the trend reasonably for the first cycle but fail to anticipate the peak after 2010.

datasets.

The fitted models are plotted against the held-out test sets which were not used in training in Fig. 3.3.

Both models generally predicted the peak around 2000 but failed for the second peak, Cycle 24. Some part of this may be due to the exceptional nature of Cycle 24; the weakness of the peak, double-peak structure, and delayed onset are notable. [16] Generally, across many train-test splits, the models do not capture the seasonal fluctuations accurately, especially when the training data is short.

Fitting the seasonal ARIMA terms is difficult due to the lack of data. The dataset only consists of four cycles: insufficient on which to train meaningful seasonal ARIMA relationships.

3.4 Fourier-ARIMA

To account for the models' difficulty in learning the seasonality, a composite model was attempted. By fitting a Fourier sum, the oscillations could capture the changing peak of the cycle. Then, an ARIMA term could be used to predict the residuals, giving the model

$$y = \text{ARIMA} + c + \sum_k (A_k \sin(2k\pi t/T) + B_k \cos(2k\pi t/T)), \quad (3.4)$$

where the sine and cosine sum allows for the phase to be shifted.

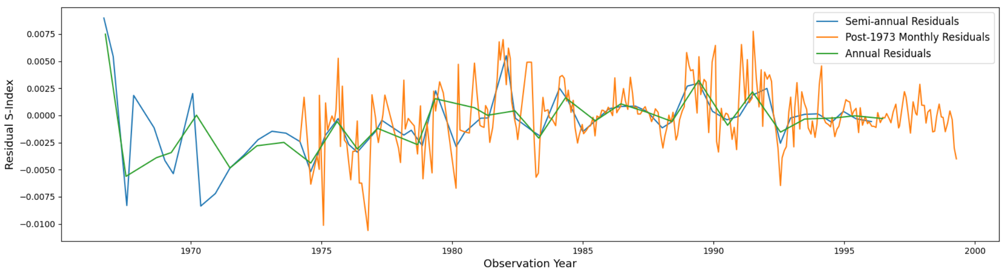

Figure 3.4: The residuals at three cadences of the Fourier fitting.

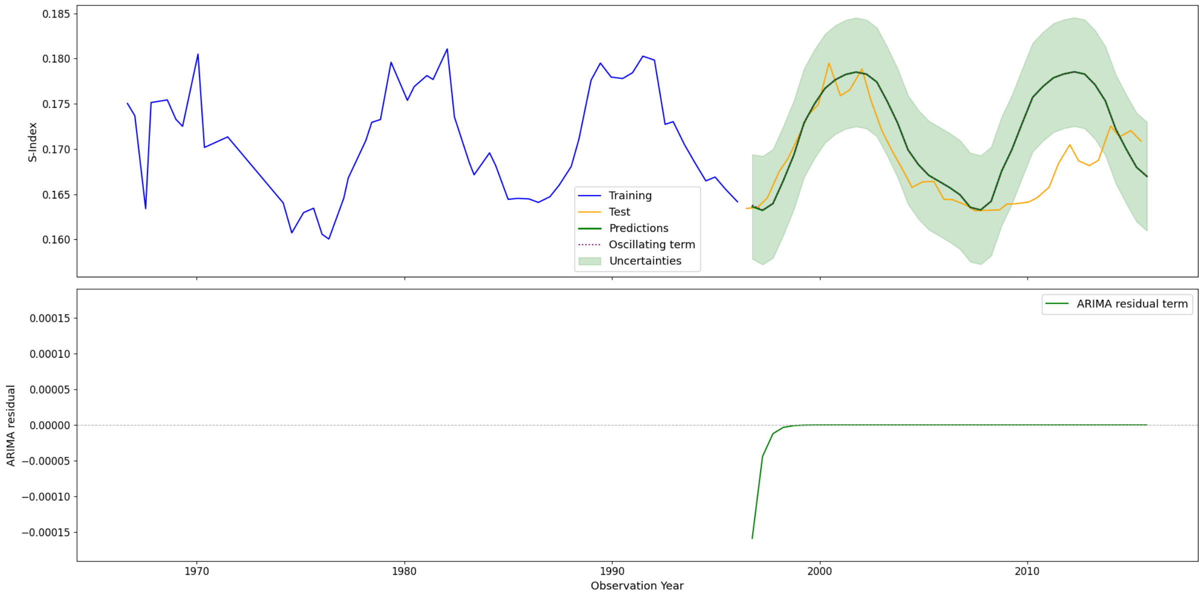

Figure 3.5: The predictions made by the Fourier-ARIMA model for the semi-annual cadence. The model fails to predict Cycle 24 because the inflexible Fourier contribution dominates.

Models with $k \in \{1, 2, 3\}$ were fitted and compared using AICc. The best fit was $k = 3$ and the residuals were taken for the annual, semi-annual, and monthly cadences, and are plotted in Fig. 3.4. Because the seasonality is handled by the Fourier portion, the monthly cadence is no longer intractable.

An example of the predictions returned by this model is plotted in Fig. 3.5. The cyclical variation is much better captured. However, when examining the contributions of each of the Fourier and ARIMA terms, the ARIMA contribution is negligible. This effectively pure-Fourier fitted model is inflexible and fails to handle changes: it struggles with predicting the anomalous Cycle 24.

This indicates that the data is overwhelmingly cyclical, and that an ARIMA component adds little once the dominant periodicity is modelled. The remaining variation, the cycle-to-cycle shape evolution, is not well-suited to a fixed-form correction using ARIMA.

The ARIMA analysis was applied to the S-Index data on the benchmark MWO datasets.

These fits struggled even more than the solar data due to fewer cycles of recorded data and more interpolated values. Values of m were chosen for these stars using the cycle lengths reported in the literature; however, this will not be possible for all stars being examined. [17]

This motivates an alternative model which can flexibly handle the periodic structure and the irregular deviations concurrently. Using a periodic kernel with a Gaussian process regression will let the model encode periodicities and, by tuning the covariances, shorter-term variations can be introduced too, without requiring the data to be downsampled or interpolated. GPR is also a Bayesian approach which should handle small datasets better by encoding the level of uncertainty compared to ARIMA's inflexible frequentist approach.

Chapter 4

Gaussian Process Regression

4.1 Gaussian Processes

Gaussian process regression (GPR) is a type of non-parametric regression that, rather than fitting a function to the data, uses the data to infer a posterior distribution over functions at unmeasured points. Given a set of measurements at $t_{1:N_{obs}}$ and their variances, the model returns a set of functions corresponding to the distribution of measurements evaluated at predicted points $t_{N_{obs}+1:N_{tot}}$ by exploiting the joint multivariate normal distribution with the covariance between the points set by a kernel function K:

$$\begin{pmatrix} f(t_1) \\ \vdots \\ f(t_{N_{obs}}) \\ \vdots \\ f(t_{N_{tot}}) \end{pmatrix} \sim \mathcal{N}(\boldsymbol{\mu}, K), \tag{4.1}$$

where the mean function μ is taken to be constant. [18] As the data is oscillatory and Fig. 3.2 indicated data stationarity, this is reasonable. This assumption will be re-examined during model construction.

The GP behaviour can be controlled by tuning the covariance matrix, which defines how predictions are affected by training points, encoding prior belief in the function's structure. Here, the kernel should encode the data's periodicity by having covariances that oscillate with the data: data is correlated with data at the corresponding point in previous cycles and anti-correlated with out-of-phase data. To adapt to the changing behaviour of each cycle, the covariance oscillation amplitude was set to decay with greater lags $\tau = \Delta t$ such that the predictions were more strongly influenced by recent data. This is called the quasi-periodic kernel, which is written as

$$k(\tau) \propto e^{-\omega_0 \tau/(2Q)} \cos(\eta \omega_0 \tau + \phi), \tag{4.2}$$

where ω_0 is the natural frequency, ϕ is the phase, and η is the ratio of the oscillation frequency and ω_0. This is analogous to simple harmonic oscillators in the underdamped regime, where the quality factor $Q > 0.5$.

4.2 Implementation

The GPRs were implemented using the `celerite2` library. [19] Computing conditional Gaussians requires inverting the $N \times N$ covariance matrix, where N is the total number of observed and predicted points: an $\mathcal{O}(N^3)$ operation. This is intractable for large datasets and high-cadence predictions. `celerite2` restricts the kernel to sums of exponentially decaying sinusoids of the form

$$k(\tau) = \sum_j a_j e^{-c_j \tau} \cos(d_j \tau) + b_j e^{-c_j \tau} \sin(d_j \tau). \tag{4.3}$$

Kernels in this form produce covariance matrices that can be inverted in $\mathcal{O}(N)$ time. This is well-suited to this project as this is the form of the quasi-periodic kernel.

Fitting quasi-periodic kernels also has the advantage of physical interpretability. Nicholson & Aigrain (2022) argue that their fitting of quasi-periodic kernels shows that fitted kernel periods recover the stellar rotation period and the decay timescales correspond to those of stellar spots. [20]

`celerite2` provides a simple harmonic oscillator term (*SHOTerm*), which implements the quasi-periodic kernel. Stellar variations often display secondary oscillations at the first harmonic of the rotation period. [21] To fit these rotation signals, `celerite2` also includes a rotation kernel (*RotTerm*) combining a *SHOTerm* and a *SHOTerm* at a harmonic.

4.3 Solar Data Manual Kernel Fitting

The GPR methods were first applied to solar data with a kernel composed of a *RotTerm* to capture rotation and an underdamped *SHOTerm* to capture the magnetic cycle. The Gleissberg cycle was not explicitly modelled at this point.

GPR models are highly flexible: good initial guesses with optimisation bounds around them were needed to force the model to converge to a physically motivated set of parameters. Without these bounds, the harmonic overtone of the rotation often grew to encompass the rotation itself. For the Sun, choosing initial guesses was straightforward because the solar rotation rate and magnetic activity cycle are well-characterised. Bounds were set to a ±10% tolerance around the known values.

The predictions this model made for various splits of solar data is plotted in Fig. 4.1. The model was generally able to predict the position of the next few extrema but failed to predict the anomalous Cycle 24 in long-term forecasting. However, in the short-term forecast (window 10), the model was able to adapt to the additional information and follow Cycle 24, supporting the choice of GPR models.

This kernel required significant manual tuning based on literature values for the expected oscillation periods, requiring confidence that the periodicity assigned to the *RotTerm* corresponded to stellar rotation. Fitting such a kernel would be difficult for other stars due to the unavailability of similar prior information. As such, it was decided to pivot to the more flexible spectrum kernel for greater generality and scalability.

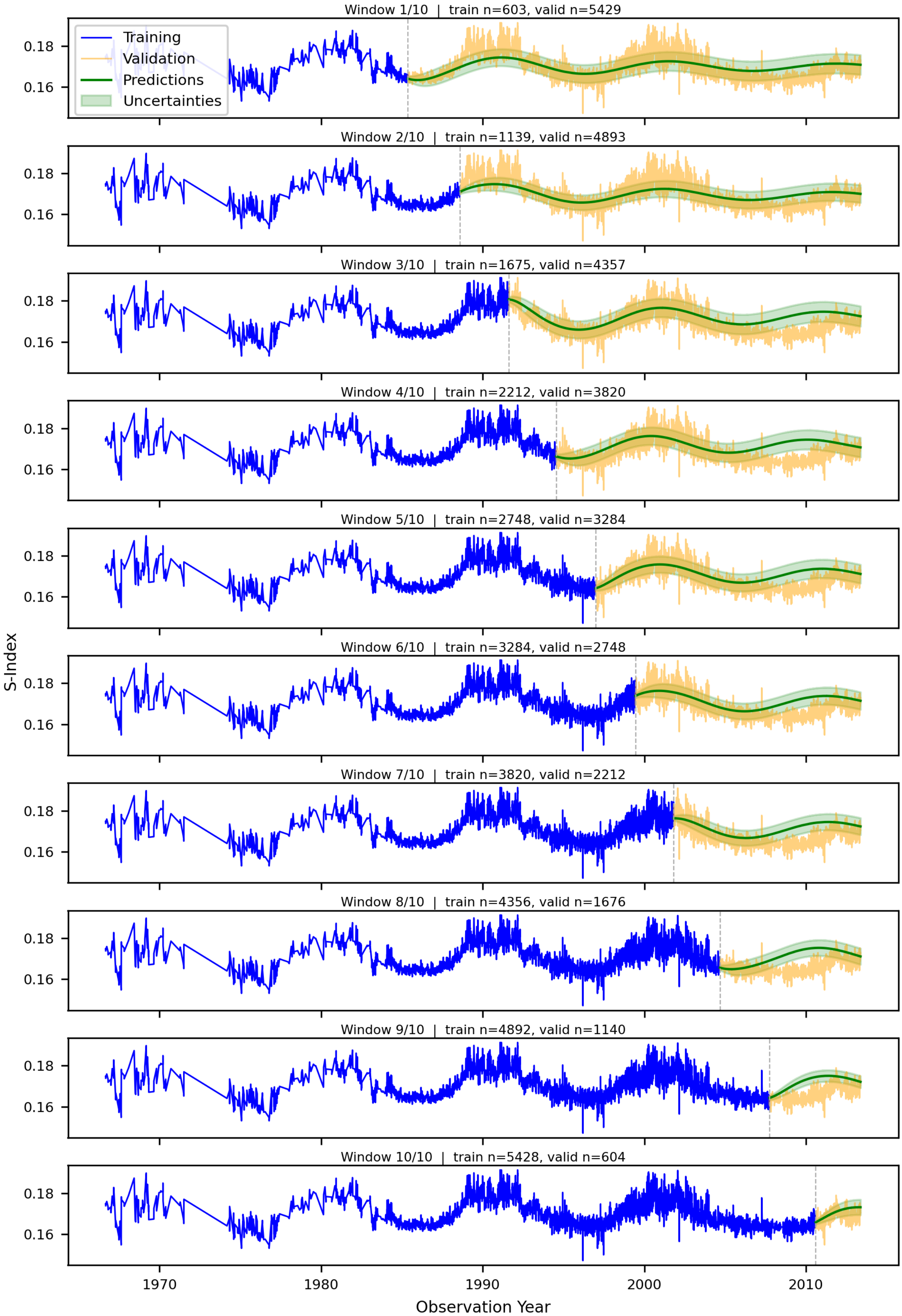

Figure 4.1: Growing window analysis for a GPR with a SHO+ROT kernel applied to the solar data. The lowest training split was 0.1 and grew linearly to the highest at 0.9. The time increments are not consistent as the sampling is irregular.

Chapter 5

Spectrum Kernel

A spectrum kernel is a summation of quasi-periodic *SHOTerm*s. Such a kernel was previously used by Gonçalves, Echer & Frigo (2020) to predict sunspot data. [6] Spectrum kernels have the form

$$k(\tau) = \sum_{i=1}^{I} \sigma_i^2 \exp\left(-\frac{|\tau|\,\omega_i}{2Q_i}\right) \cos\left(\tau\omega_i\right), \tag{5.1}$$

where σ_i^2 is the amplitude, Q_i is the quality factor corresponding to the decay timescale, and ω_i is the term's angular frequency controlling the oscillation period.

This spectrum kernel instead fits to whatever signal has the strongest effect on the data rather than imposing requirements on what harmonics ought to be present. To physically motivate each spectrum kernel mode and to prevent their periods from converging to the dominant oscillation, three modes with distinct period ranges were considered, each corresponding to the periodicity of one of the stellar activity cycles described in section 1.3. The three modes are:

1. the *short* cycle, corresponding to periods where the signal could be interpreted as a rotation signal;

2. the *mid* cycle, corresponding to the range of periods of stellar magnetic activity cycles;

3. the *long* cycle, corresponding to the long-term variability in the magnetic activity cycle, such as the solar Gleissberg cycle.

The spectrum kernel defined in Eq. 5.1 models the long cycle additively. However, the Gleissberg cycle modulates the magnetic cycle amplitude: a multiplicative effect. [9] An alternative model was considered, in which the long cycle was predicted separately on the amplitude envelope and applied multiplicatively to the additive short and mid cycles.

However, the additive model was used instead as a long-term modulation could be more simply captured by a long-term cycle adding to the magnitude as required. An additive long-term cycle is also much more flexible. Considering that the timespan of the MWO data is shorter than that during which a long-term cycle can be expected to vary measurably, a Gleissberg-like cycle would be difficult to detect. It is, however, conceivable that over the decades of observations, there may be systematic instrumental drift. Drift could also be handled by the long-cycle term by fitting a very long-period oscillation. As discussed in section 4.1, GPR modelling often treats the mean function as constant. By accounting for the instrumental drift, the additive long-term cycle ensures the validity of this key assumption.

This kernel enables easy model variation and comparison. Not all three modes are present in all data. By varying the number of modes and the cycles to which they correspond, the kernel can be adapted flexibly. The best kernel mode combination was selected by model comparison on a validation set. By eliminating the requirement for manual kernel tuning for each dataset, this process is designed to generalise well to new datasets. Then, the selected model was retrained on the training and validation data for further predictions.

Each *SHOTerm* defined by `celerite2` has three parameters: σ_i, ρ_i (related to ω_i), and Q_i. The quality factor is the decay timescale of the covariance in oscillation cycles, interpreted as the number of cycles over which the mode retains memory. To ensure physical fits for all stars, the priors (initial optimisation states) and the bounds for each mode had to be defined. The training data was used to define these without manual intervention so that the process would be scalable to datasets of many stars.

The prior selection and model fitting analysis pipeline will be described and then validated in this chapter.

5.1 Defining Priors

5.1.1 Sigma Priors

The σ values were motivated by the amplitude of the stellar data variability; the training dataset variance was taken as a reference value. To reflect ignorance in each mode's relative strength in different stars, each of the I prior magnitudes was defined equally

$$\sigma_i = \frac{\sigma_{\text{train}}}{\sqrt{I}}. \tag{5.2}$$

5.1.2 Q Priors

Quasi-periodic oscillations are characterised by Q values above 0.5, providing an easy lower bound. For most MWO stars, observations were only made between 1966 and 1995. For the mid and long cycles, which are periodic on the order of one and many decades respectively, the number of cycles of training data available is therefore constrained by the span of the observations, not Q. The short cycle is in the other regime: there are many cycles of data available.

There is little information on how strongly each type of cycle's covariance should decay at increasing lags. As such, the Q priors were randomised to be between 0.5 and 1 but were given wide optimisation bounds.

5.1.3 Rho Priors

The period priors were extracted by examining the periodicities apparent in power-spectra taken of the training data.

A Lomb-Scargle periodogram (LSP) is a method for detecting periodic signals in unevenly sampled data by fitting sinusoids across a range of trial periods. By calculating how much each fit reduces the scatter of the residuals, how well a sinusoid at each trial period explains the data is quantified. Periodicities of fits removing much scatter correspond to periodic variations in the data. [4]

LSPs output dimensionless power across the sampled periods. The power-spectral peaks can then be used to inform the ρ priors.

The LSP was evaluated on a logarithmic period grid from 0.1 days to 100 years with 100,000 points. The logarithmic grid allows for periodicities across all the cycle types to be detected simultaneously while a linear grid was unsuitable because undersampling at the shorter periods leads to proportionally larger relative error due to aliasing.

The significance of each LSP peak was determined by the false alarm probability (FAP). This calculates, under a white noise assumption, the level of power reached due to random fluctuations corresponding to a p-value.

The naïve application of an LSP to the raw data returns a periodogram dominated by one peak. This is because the LSPs normalise power relative to the data's total variance, accentuating the power of the dominant peak. An example of this first-order LSP calculated on the solar time series is plotted in Fig. 5.1.

To increase the dynamic range, the dominant signal variance was removed by fitting the sinusoid and calculating a new LSP on the residuals. Peaks on the residual LSPs were identified using `scipy.signal.find_peaks`. In each iteration, the most powerful peak was checked against alias, window function, and SNR criteria; if it passed, it was accepted as physical. Then the accepted periods were fitted simultaneously to create new residuals.

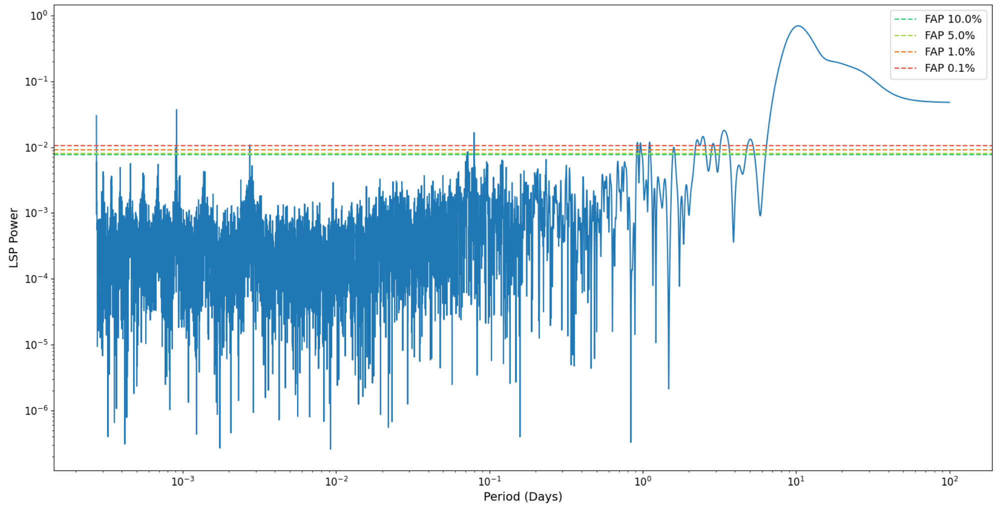

Figure 5.1: The first-order LSP taken on the solar data is truncated to a maximum period of 100 days and plotted in log-log to emphasise the shorter periods and lower magnitude peaks. The magnitudes of several false alarm probabilities are plotted. There are several peaks above the 0.1% FAP, indicating strong evidence of cyclical variations.

This was repeated until there were no more signals above the 0.1% FAP threshold.

Aliases are harmonics of physical signals, which become indistinguishable at low sampling cadences. Signals within 5% of harmonic multiples of previously accepted periods were discarded. Window functions correspond to the observing cadence. Candidate periods within 5% of common cadences were removed.

Each peak's signal-to-noise ratio (SNR) was checked to ensure that the found peaks were significant. SNR was defined as the ratio between the peak power of a Gaussian fit on the LSP to the residual LSP's standard deviation after subtracting the fit. If the SNR was greater than an arbitrary threshold of four, the period being processed was accepted and added to the list of periods. The SNR check was performed last as it was the most computationally intensive, ensuring that it would only be run if the signal would not be otherwise disqualified. If a period failed a check, the next strongest peak in that LSP was checked until a period was accepted or there were no more significant peaks, ending the loop. An example of this process is plotted in Fig. 5.2.

This process yields a list of periods. Each was extracted from a differently normalised LSP. As such, to compare their strengths, the powers at those periods were evaluated on the first-order LSP. These were then processed into physically motivated priors for each mode by classifying them into bounds corresponding to the targeted cycles.

Suárez Mascareño et al. (2016; henceforth SM2016) studied the rotation and magnetic cycle periodicities across 176 stars, reporting the mean periods and variances for different stellar classifications. [8] The most extreme reported three-standard-deviation rotation

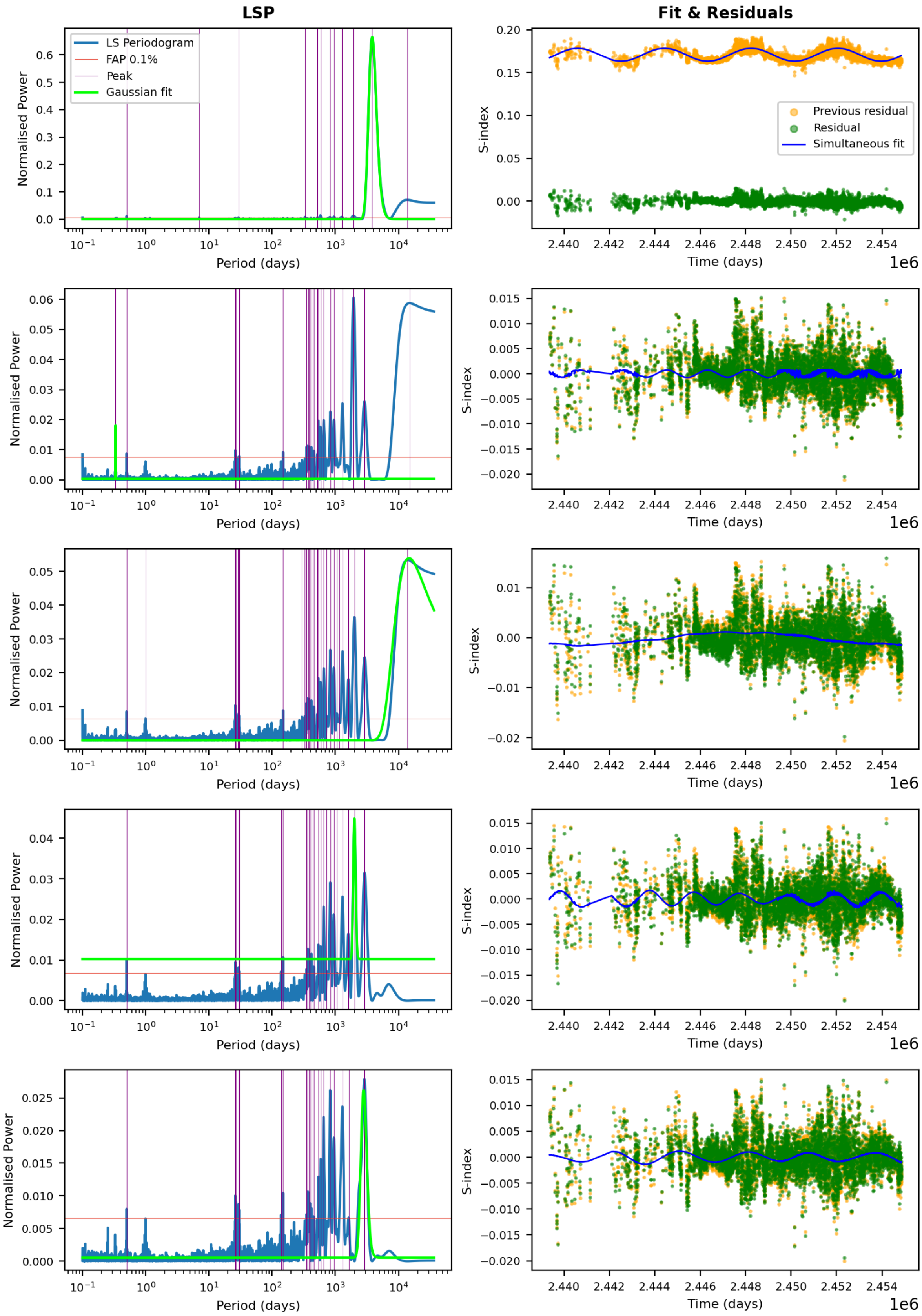

Figure 5.2: The first five iterations of the peak-finding loop with each iteration's LSP plotted in the left column and the residuals plotted in the right column. Gaussians fitted for the SNR check are shown in green in the right plot and the peaks identified are indicated with purple. Only peaks more significant than FAP 0.1%, indicated by a horizontal red line, are considered.

cycle length, 250 days, was chosen as a safe upper threshold for the classification of a peak into the short cycle corresponding to stellar rotation. This is comfortably above the longest observed rotation cycle of 160 days.

The mid cycle threshold was similarly defined to be 25 years. Any period longer than this was assigned to the long cycle.

When a cycle had more than one period, the most powerful one was chosen. If a cycle had no detected period, the best prior was estimated. SM2016 found a correlation between the rotation and magnetic cycle lengths for FGK stars. So, with a prior on one of the short or mid cycles, a prior on the other can be estimated using

$$\log_{10}(P_{\text{short}}) = \frac{1}{1-a}\left(\log_{10}(P_{\text{mid}}) - c\right), \tag{5.3}$$

where a is the slope parameter. Values for c were calculated for each star type based on the mean magnetic and rotation cycle lengths reported in SM2016.

If no mid or short prior was available, or the star was not FGK, the prior was estimated by taking the reported mean values for the short and mid cycles for that star type. Due to the difficulty in observing the long cycles in short data, statistics on the average length of such variations are not available. In the absence of better data, 100 years was used as an estimated prior for the long cycle when no long-term trend is detected by the LSP analysis.

HD81809 (G-type)		
Cycle Type	**Found**	**Literature**
Short (rotation)	40.25 days	39.3 days
Mid (activity)	2991.66 days (8.20 yr)	8.2 years
Long	— (defaulted to 100 yr)	—
HD160346 (K-type)		
Cycle Type	**Found**	**Literature**
Short (rotation)	35.14 days	35.3 days
Mid (activity)	2505.48 days (6.86 yr)	7.0 years
Long	23064.62 days (63.19 yr)	—
HD201091 (K-type)		
Cycle Type	**Found**	**Literature**
Short (rotation)	36.97 days	34.0 days
Mid (activity)	2730.68 days (7.48 yr)	7.3 years
Long	12043.36 days (33.00 yr)	—

Table 5.1: Comparison of found cycle periods with literature values for HD81809, HD160346, and HD201091. The priors found for these benchmark datasets lined up well with the literature.

This analysis was tested on the benchmark stars plotted in Fig. 2.1, which have rotation and magnetic activity cycle periods in the literature. [17] The results are summarised in Table 5.1. Values for the short and mid cycle periods found using the LSP analysis applied to the well-sampled benchmark stars agree well with reference values.

5.2 Defining Optimisation Bounds

Due to the flexibility of GPRs, bounds on the priors had to be set to prevent each mode from wandering into the domain of another mode. To reflect the different levels of certainty about the ρ priors calculated by the three different methods, bounds of different widths were set for each to allow greater variation for more uncertain priors.

Direct LSP observations were the most reliable, and the optimisation of modes using them was constrained to a relative bound of $\pm 10\%$. To reflect the increased uncertainty from the indirect prior calculation using the SM2016 relation, optimisations using these priors had a relative constraint of $\pm 20\%$. The population mean method was the most unreliable and had the largest bounds. As the standard deviations are also reported in SM2016, the bounds were set to be one standard deviation from the mean. All the reported standard deviations were large relative to their means; as such, more than one standard deviation was too wide.

The long-cycle mean estimate of 100 years is especially uncertain as this is an order-of-magnitude estimate based on a single star: the Sun. Long-term trends affecting the instrument are also unknown, so it is important for the long cycle mode to be flexible. Reflecting this uncertainty, long-cycle mean estimates were given ± 50-year bounds.

If this created overlapping optimisation bounds for different modes, degenerate solutions, where the periods of several modes collapse to the same value, would be penalised during the model comparison as they added no information while using more parameters.

σ was given a lower bound of 10^{-4} and an upper bound of five times the training standard deviation. Only a relatively loose bound is required because overly large σs will be penalised by the loss function in training and overly small σs will be penalised by model selection for degeneracy.

The Q bounds were similarly loose as they only had to be above 0.5 to force under-damped *SHOTerm*s. However, larger Qs produce signals that are coherent across longer forecast horizons. To bias the models towards the ability to sustain multi-cycle forecast horizons, the lower bound was set higher, at 1. The upper bound was set to 10 as an arbitrary loose limit.

5.3 GPR Optimisation

The parameter space is non-negative with parameters that can vary by orders of magnitude, motivating parameter fitting in log-space. Parameter optimisation was performed by negative log-likelihood minimisation, which is given up to an additive constant by

$$\mathrm{NLL} = \frac{1}{2}\mathbf{y}^T K^{-1}\mathbf{y} + \frac{1}{2}\log|K|. \tag{5.4}$$

The first term reflects how well the kernel describes the training data. The second term is an Occam penalty that penalises overly flexible covariance structures.

An L-BFGS-B optimiser was used because it handles the bounds naturally. It is a gradient-based optimisation method, making it well-suited for use with `celerite2`, which calculates likelihoods analytically, returning smooth likelihood functions. Due to the deterministic approach, L-BFGS-B optimisers can get stuck in local minima. To mitigate this, the optimisation was run 50 times with Gaussian noise around the priors and the optimisation run with the lowest NLL was taken to be the overall result.

5.4 Model Comparison

Using the priors and bounds, spectrum kernels were constructed from every combination of the short, mid, and long modes, with the number of modes $I \in \{1, 2, 3\}$. Another type of model that corresponds to a non-oscillatory, constant activity star was considered by having a *SHOTerm* kernel in the overdamped regime ($Q < 0.5$). These kernels were optimised according to the procedure in section 5.3.

It was noted that models including the mid mode tended to predict the cycles best: purely short-mode kernels fitted to noise and long-mode kernels are intended to modify the mid mode. As such, any combination of modes without a mid term was not considered.

The constant kernel was also troublesome: the model frequently defaulted to it in cases where the data span was short, even when the S-Index was clearly oscillatory. Rather than abandoning forecasting when the data was limited, as is frequently the case, it is preferable to accept a less accurate forecast. Accordingly, the constant kernel was removed from consideration. Instead, inactive stars were handled by the other kernels with suppressed amplitudes.

Irregular sampling also caused issues: observations were taken in bursts due to observing seasons. Since stellar activity varies on timescales much longer than any single burst, the data can appear locally constant within these clusters. When the training data ends during such a burst, the kernel's downweighting of past data means predictions fail to oscillate. To mitigate this, data was downsampled to a maximum of one observation per night.

The kernel combinations were initially compared using their validation set negative log-predictive densities (NLPD), a metric that evaluates the probability mass of the predicted distribution at the validation points. NLPD is harsh on overconfident models because the logarithm term explodes if the model assigns near-zero density to the validation values, penalising vagueness less than misplaced but sharp features. [22] As such, comparison by NLPD pushes the selection towards models with weak oscillations and large uncertainties. This biased the pipeline away from models with multiple kernel modes, often selecting the most basic kernel with only the mid mode in order to suppress oscillations, preventing the full characterisation of the variation. NLPD comparison also tended to select the model with the largest uncertainties. This is undesirable: it is more important for forecasts to have prominent oscillations than for predictive uncertainties to encompass the observations since the priority is being able to clearly identify the best time for observations. Therefore, the kernel combinations were instead compared using the continuous ranked probability score (CRPS).

CRPS integrates the squared difference between the predicted and empirical CDFs of the validation data. By considering the full distribution shape, not just one point, the metric also rewards nearly correct predictions. [22] CRPS comparison instead favoured more strongly oscillatory models, with varying numbers of modes, that better matched the data's oscillations. As such, the kernel combination with the lowest CRPS was chosen.

5.5 Parameter MCMC

After model comparison, the training and validation sets were combined into the retraining set; the test set was still held out for reporting performance. The GPR naturally handles aleatoric uncertainties through the diagonal variance terms in the covariance matrix. [18] Epistemic uncertainty in the calibrated parameters and between the kernel combinations must be considered to fully characterise uncertainty. Ideally, a reversible-jump MCMC (RJMCMC) would be implemented to characterise both simultaneously. However, the design of a sampler that can jump between kernel combinations while satisfying detailed balance is a substantial undertaking beyond the scope of this work, and is left as an extension. The uncertainty on the kernel parameters was found by performing MCMC sampling with the `emcee` package. [23]

Normally distributed perturbations around the selected kernel combination's optimised parameters were set as the initial positions for 32 walkers. The posterior was defined by a uniform sampling prior and the `celerite2` log-likelihood. Burn-in was removed and the samples were thinned. Posterior parameter samples were substituted into the kernel functions to generate predictions. The prediction was from the mean of the posterior samples; the prediction uncertainty was defined as a sum of the variance between sample predictions (epistemic contribution) and the mean variance within each sample's predictive distribution (aleatoric contribution).

5.6 Evaluation

The models were retrained on the retraining dataset and evaluated on test data. Due to the irregular sampling of the test data, it had to be interpolated to provide an accurate ground truth. Initially, a moving average was tested but assigning a principled window span parameter was difficult. The moving average also exhibited large discontinuities due to the sampling.

Under the quasi-periodic assumption, the periodicity should not drift significantly in the test dataset. As such, the ground truth was defined as a three-mode Fourier series, matching the three GPR modes. To make use of all available data, the series was fitted on the full dataset but weighted to reflect the quasi-periodicity. The retraining data weights decayed exponentially with characteristic timescale equal to the mid-cycle prior for that model.

A Gaussian noise likelihood was defined and the Fourier series parameters were sampled such that the uncertainty could be ascertained. An example of a growing window analysis performed using these methods is plotted in Fig. 5.3.

Since the forecasts are intended to inform observation scheduling, the best time to observe a star within a lookahead window was chosen as a flexible metric to report. The times of the next minima were not reported because extracting good minima times from the predictions was challenging. This is because, at short horizons, predictions are heavily weighted by often noisy data, causing boundary oscillations. The prediction time-grid for the first year was defined with a coarser sampling to mitigate the ringing. However, there are often still noisy minima which any minima-finding algorithm would falsely identify as the best observation time.

Lookahead (yr)	**MAE (yr)**
1	0.251
2	0.512
3	0.667
4	0.916
5	1.066

Table 5.2: Mean absolute error of the predicted optimal observation times as a function of lookahead window.

To validate the analysis, it was performed for 25 splits for each of the benchmark datasets. The errors between the forecasted and actual best observation times were noted

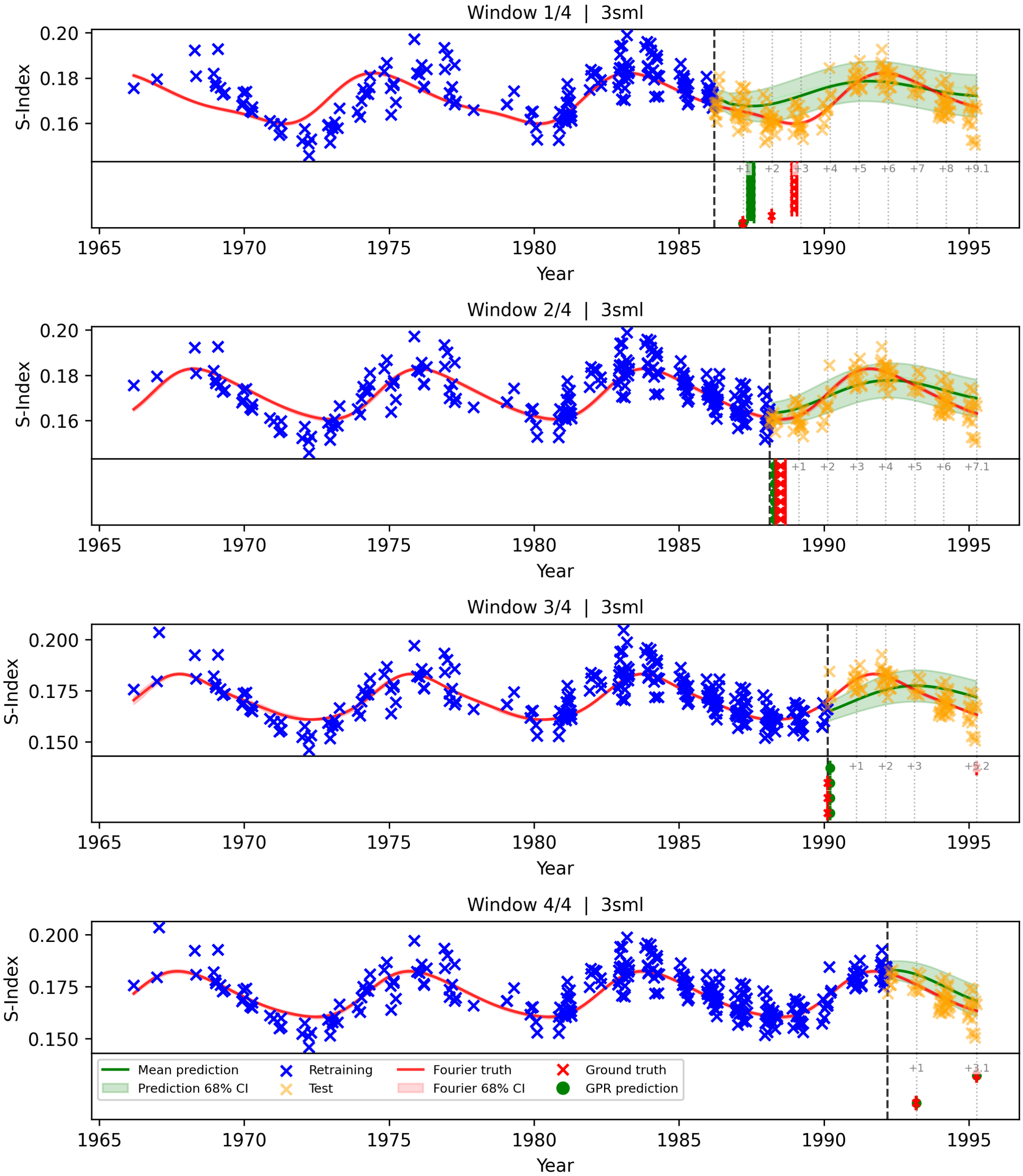

Figure 5.3: The time series predictions for HD81809 for four different splits. The top plot is the time series where the training and validation data used to train the final model are plotted in blue, the predictions of the model and uncertainty thereof in green, the test set data in yellow, and the fitted ground truth in red. The bottom plot for each window indicates the predicted best observation times for each lookahead window in green (furthest lookahead at the top) and the ground truth best observation time in red. Vertical lines spanning both panels indicate the length of the lookahead windows considered. Uncertainties were calculated by the method described in section 5.5. They generally underestimate uncertainty because the epistemic uncertainty between kernels has not yet been characterised. Errors between predicted and actual best observation times were generally much smaller than the timescale of the variations.

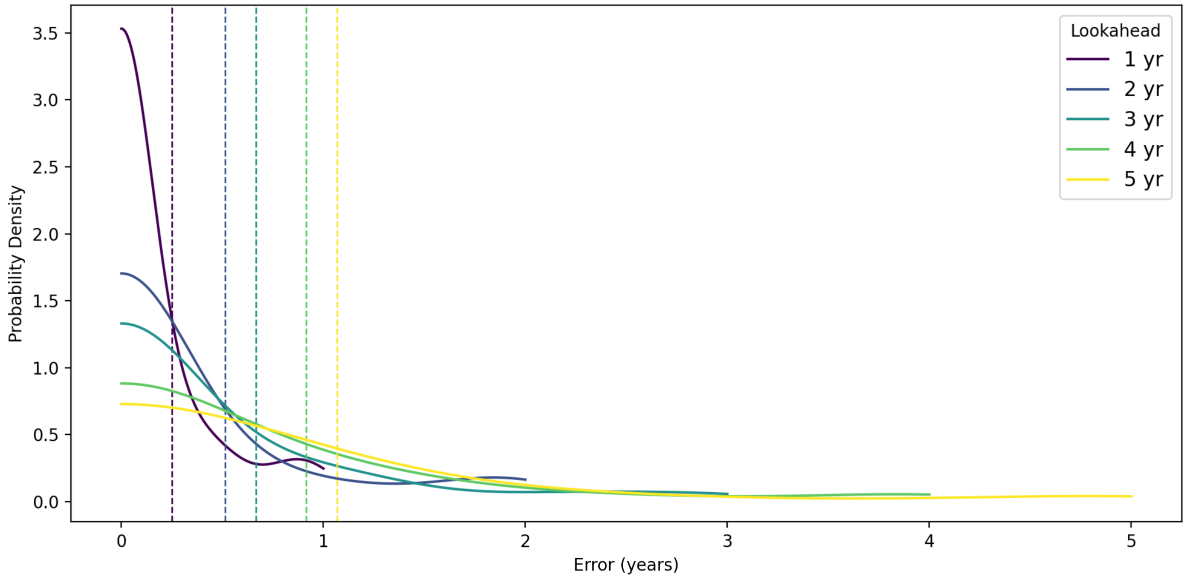

Figure 5.4: The error distribution for annual lookahead windows for the first five years calculated on the benchmark datasets. The mean of each distribution is plotted as a vertical dashed line. The mean error magnitude grows for larger lookahead windows, reflecting the difficulty in producing a long-term forecast based purely on the time series data.

for various lookahead windows. The distribution of the magnitude of the errors was examined by plotting a normalised kernel density estimate (KDE) to get the smooth distribution in Fig. 5.4. The mean errors of the distributions in Fig. 5.4 are given in Table 5.2. These show that, while errors grew with lookahead window length, they remained below or approximately one year. This is encouraging given that the variations of interest occur on multi-year timescales.

This analysis was attempted on a much larger set of oscillatory MWO stars identified by Isaacson et al., but repeated HPC failures prevented completion. [24]

The effect of seasonal observation cadence on pipeline performance was quantified to inform how future S-Index observations should be made. This was done by simulating a population of 100 stars according to the SM2016 population survey and assessing the performance when the data was sampled at different rates. Star types were drawn from a categorical distribution. The rotation and magnetic cycle periods were then drawn according to the normal distributions defined in SM2016, while the long-term envelope oscillation period was randomised between 50 and 150 years.

There are no population data available regarding the relative amplitudes of the modes, leaving this a matter of scientific uncertainty. Solar sunspot data show that sunspot counts vary by up to a factor of two times peak-to-peak. [25] As such, long-cycle amplitude was drawn between one (no oscillation) and two. The rotation and cycle amplitudes were drawn from a KDE defined by the stars for which SM2016 reported both amplitudes.

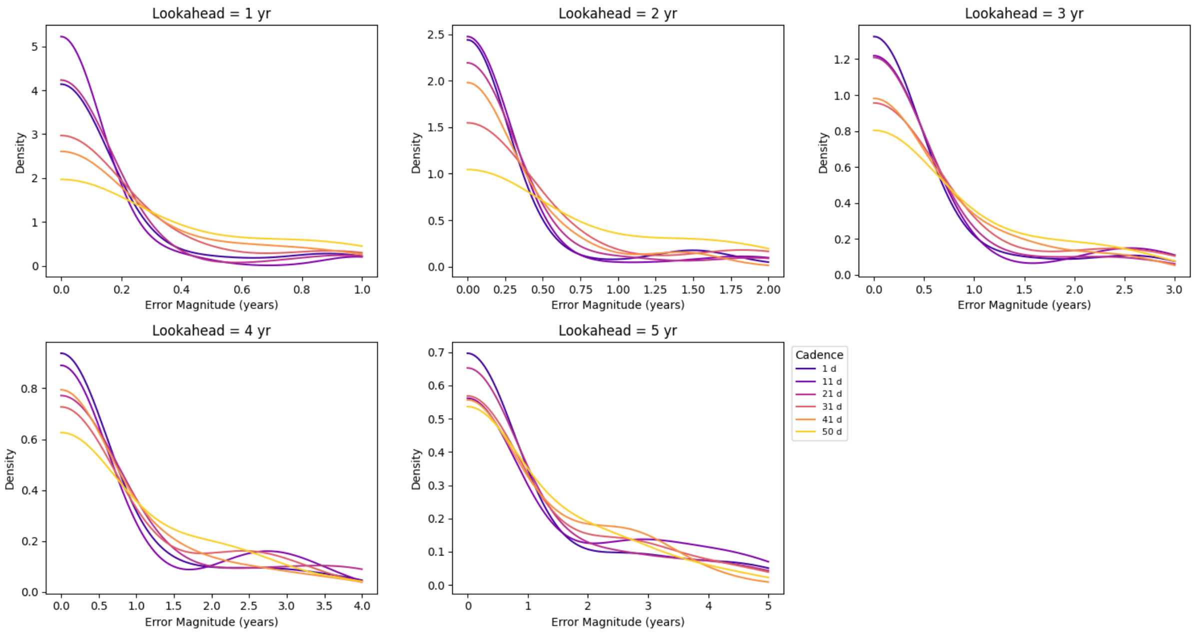

Figure 5.5: The distribution of lookahead errors for various sampling cadences at various lookahead windows calculated on 15 simulated stars. Mean errors increase as sampling becomes sparser as expected. The distributions are not truly Gaussian, growing more irregular at higher lookaheads. This indicates that there is some sampling noise due to the low-data regime with only 15 datasets. The decrease in performance with cadence is not strictly monotonic at lower cadences; this is likely due to the noise.

These artificial stars were observed seasonally with observing periods sampled at six cadences between 1 and 50 days. Noise was added to the data with a relative magnitude of 2.5%. The lookahead window analysis was performed on these simulated datasets and the results are plotted in Fig. 5.5. Due to the HPC failure, the analysis could only be performed with 15 out of the 100 simulated stars per cadence. For shorter lookaheads, the high cadences achieve the lowest errors, with performance decreasing for lower cadences. The effect is less pronounced at longer lookaheads as performance converges, reflecting the difficulty in making long-term predictions. These preliminary results suggest that future observing campaigns intended for stellar cycle forecasting should choose the sampling cadence for each star based on the desired forecasting window. The full analysis should be performed when more compute becomes available to provide a quantitative reference to support this.

Chapter 6

Conclusion

Stellar activity injects non-planetary noise into exoplanet searches, with noise magnitude scaling with a star's activity cycle. As such, exoplanet detection sensitivity can be maximised by scheduling observations of stars during periods of lower activity. To this end, this work developed a scalable pipeline for forecasting stellar activity cycles from S-Index data.

An ARIMA baseline was constructed but was fundamentally limited by the short span of MWO data: there were too few recorded cycles. A Fourier-ARIMA composite model showed the data is overwhelmingly cyclical with negligible ARIMA contribution. However, the Fourier fit was too rigid to capture cycle-to-cycle variations, motivating the GPR spectrum-kernel approach, which flexibly handles periodic structure alongside irregular deviations.

Spectrum kernels reproduced the oscillations of the benchmark stars well. Reporting "best time to observe within a lookahead window" gave prediction errors below approximately one year for lookahead windows up to five years, which is encouraging since the variations of interest occur over several years.

The work completed was severely limited by the prolonged HPC failure. As such, the pipeline evaluation was limited to the well-sampled benchmark stars; the large-scale active star dataset run could not be completed. This limits the empirical evidence for pipeline scalability, but not the argument for it: the pipeline was optimised for large runs. The automation of the prior generation and model comparison is designed to eliminate the need for manual tuning in datasets that can have thousands of stars. Furthermore, because of efficient kernel design, the GPR pipeline's complexity scales linearly rather than cubically with the number of data points, supporting its application to large datasets.

Similarly, due to HPC failure, the synthetic-population data-quality study was run on only 15 stars per cadence. While the trends are noisier than ideal, they already show a consistent trend of degrading performance with sparser cadence, which can help inform observing strategies.

The most immediate priority for future work is to complete the full-population run and the cadence study, to more strongly verify pipeline performance. An up-to-date S-Index dataset should then be assembled to support an operational forecast using the pipeline. Furthermore, an RJMCMC sampler should be implemented to fully characterise the uncertainty in kernel selection. The cadence analysis results should be validated against sparsely sampled and downsampled stars, then used to recommend future observing strategies.

Overall, this work has demonstrated that, for the benchmark stars tested, GPR can produce reliable activity forecasts for horizons up to five years from S-Index data, providing a foundation for the public forecasts Sairam & Triaud called for.

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Appendix A

Autogeneration Statement

LLMs, specifically Claude Haiku 4.5, Claude Sonnet 5, and Claude Opus 4.8, were used to support this work. It was used to:

- Help debug code when human attempts failed.
- Improve low efficiency code: "how could this code be made more efficient?"
- Help with some of the more esoteric Matplotlib functions: "how do I make this plot look like xyz?"
- Help rephrase unwieldy sentences: "how can I make this sentence flow better?"
- Generate the repository structure section in the README.md.
- Wrap code for use in HPC: "how could I write this for HPC use?"
- Identify test cases for continuous integration: "what should I test in xyz function and how could I do that?"

I would like to emphasise that the analysis and thought processes are **my own**. Model outputs were always reviewed and checked against literature. No generated response was incorporated into the work without thorough verification.